

Rebuilding Section 4 of the Three-Mode NLS Study

Preregistered stationary-density and held-out generator-mode validation

14 September 2026

Abstract

We rebuilt the short-chain analysis around a frozen validation hierarchy. Complete trajectories were split into training, validation, and blind-test sets; the equilibrium Gibbs state was used to calibrate stationary-density accuracy before any nonequilibrium fit; and the generator-mode analysis was performed independently on held-out trajectories. The stationary-density branch gives a decisive negative result: none of 18 normalized neural fits passes the preregistered equilibrium Fokker–Planck residual gates, so no NESS density is trained and no full five-dimensional numerical-solution claim is made. The independent spectral branch does validate a slow EDMD-visible real mode and a faster damped oscillation, but the existing common-plateau gate does not permit identifying the former as the unique spectral gap.

1 Frozen design and inputs

The protocol was frozen before model fitting. Its SHA-256 is

0513a546ba80ea0bd8abbbb990c93b74bc416e83732701941429bea391ae1b70.

For each bath condition and timestep, 64 independent controlled trajectories were partitioned by a deterministic trajectory hash into 32 training, 16 validation, and 16 blind-test streams. Snapshots from one trajectory never cross a split. Density and mode analyses used independent split salts.

The controlled integrator uses double precision, fixed timesteps, standard trigonometric functions, implicit-midpoint Hamiltonian evolution, Cartesian bath updates, and no action floor or projection. A direct float64 state replay was required because two historical float32 sum/difference records cancelled to zero when reconstructing an endpoint action. The replay preserved all 256 trajectories: the maximum discrepancy in the old saved observables was 1.14×10^{-13} .

Table 1: Primary split and operator gates.

Gate	Result	Evidence
Fine driven blind ESS	PASS	11,616 (threshold 5,000)
Fine equilibrium blind ESS	PASS	13,860 (threshold 5,000)
Reduced drift audit	PASS	max abs. error 2.27×10^{-13}
Reduced covariance audit	PASS	max abs. error 2.91×10^{-11}
Gibbs adjoint identity	PASS	RMS relative residual 1.51×10^{-14}
Normalized-density AD smoke test	PASS	finite, normalized, differentiable

An initial weak-identity family contained six bare angular probes whose generator terms are not square-integrable at zero action. This domain error was identified and documented before density fitting. The corrected family contains 18 regular functions. A 2,000-replicate whole-stream equilibrium bootstrap set the simultaneous threshold to 3.748 standard errors; the direct equilibrium and driven blind trajectory checks had zero failures.

2 Stationary-density known-answer test

In the reduced coordinates

$$X = (I_1, I_2, I_3, \theta_1, \theta_3), \quad E = \frac{1}{2}M^2 - \frac{1}{4} \sum_{j=1}^3 I_j^2 + I_1 I_2 \cos \theta_1 + I_2 I_3 \cos \theta_3,$$

the equal-temperature invariant density is $\rho_{\text{eq}}(X) \propto \exp[-E(X)/T]$. We used a normalized conditional mixture model in $u_j = \log I_j$: a Gaussian mixture for u and conditional von Mises factors for the two periodic angles, with the exact Jacobian $\rho(I, \theta) = p(u, \theta)/(I_1 I_2 I_3)$.

Six architecture/penalty families and three seeds per family were declared in advance. A family was admissible only if every seed had Gibbs slope in $[0.95, 1.05]$, median $|L^\dagger \rho / \rho| \leq 0.20$, and 90th percentile at most 1.00 on the equilibrium validation split.

Verdict. FAIL. The best median residual is 1.133, 5.7 times the declared 0.20 gate; the best p90 residual is 3.705, 3.7 times its gate. The selector therefore returned `NO_ADMISSIBLE_CANDIDATE`. Following the registered stop rule, we did not inspect the equilibrium density blind split and did not train a driven density. The present model family does not support a claim that the five-dimensional stationary Fokker–Planck equation has been numerically solved.

3 Implication for the stabilization subsection

The density failure does not invalidate the controlled stationary trajectories: their operator, ESS, and regular weak-identity checks pass. It does prevent us from assigning quantitative probability to high- and low-energy regions using this neural density. Thus a density-based stabilization mechanism is not evaluated. It should be revisited only with either a new model class that independently passes the same equilibrium calibration, or with directly estimable conditional balance observables and separately frozen controls.

4 Held-out generator modes

The mode branch was run independently. EDMD operators were fitted on 32 streams, selected on 16 validation streams, frozen, and evaluated once on 16 test streams. The slow real mode used $\tau = 0.5$. The targeted exchange-odd complex mode used $\tau = 0.05$, safely below its principal-log Nyquist limit.

The real-mode blind intervals overlap between timesteps for each bath condition. The fine-step frequencies agree with independent fits (5.500 driven and 5.264 equilibrium), and the damped model is strongly preferred by AIC. These are held-out validations of modes visible to the frozen EDMD dictionary.

The slow-mode weight ordering is stable: I_2 (0.47–0.50), $I_1 + I_3$ (0.28–0.30), the cosine sum (0.20–0.22), and $\cos \theta_3$ (0.106–0.124). This explains the different visibility of the slow mode in raw observables. Nevertheless, the preregistered common-autocorrelation-plateau gate remains false. The validated real mode must therefore be called an *EDMD-visible slow mode*, not the unique spectral gap of the full generator.

5 Audit trail and claim boundary

All analysis-time repairs are documented before the affected rerun: float64 direct-state output, serialization-only model saving, selector filtering of preserved failed-save directories, generator-domain correction of the weak family, and the deterministic modal-weight read. None changes a scientific threshold or selects a result by closeness to a target.

Permitted claim. The controlled three-mode trajectories support a held-out EDMD-visible slow real mode and a faster damped oscillation. The tested normalized neural density family fails the equilibrium Gibbs/Fokker–Planck known-answer calibration; therefore the full NESS density, density-derived stabilization mechanism, and unique spectral gap remain unresolved.

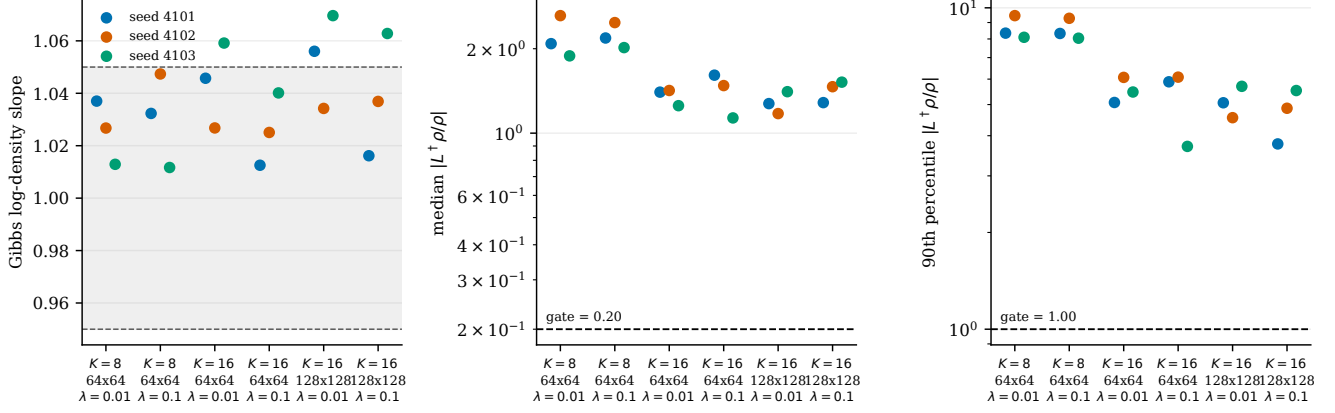


Figure 1: Complete equilibrium validation matrix. The slope alone can look reasonable while the pointwise stationary-equation residual fails by a large factor. Each point is one independently trained seed.

Table 2: Equilibrium validation summaries; each row contains three seeds.

K	widths	λ_{FP}	median NLL	Gibbs slope	centered RMSE	median residual	p90 residual	admissible
8	64x64	0.01	7.9010	[1.0129,1.0370]	[0.4558,0.5471]	[1.886,2.624]	[8.096,9.456]	no
8	64x64	0.10	7.9487	[1.0117,1.0474]	[0.5454,0.5954]	[2.018,2.478]	[8.044,9.277]	no
16	64x64	0.01	7.8255	[1.0268,1.0592]	[0.3034,0.3520]	[1.253,1.419]	[5.075,6.075]	no
16	64x64	0.10	7.9112	[1.0126,1.0401]	[0.4646,0.5319]	[1.133,1.610]	[3.705,6.085]	no
16	128x128	0.01	7.8291	[1.0342,1.0696]	[0.3020,0.3533]	[1.174,1.406]	[4.552,5.698]	no
16	128x128	0.10	7.9494	[1.0162,1.0628]	[0.4511,0.5860]	[1.285,1.520]	[3.772,5.529]	no

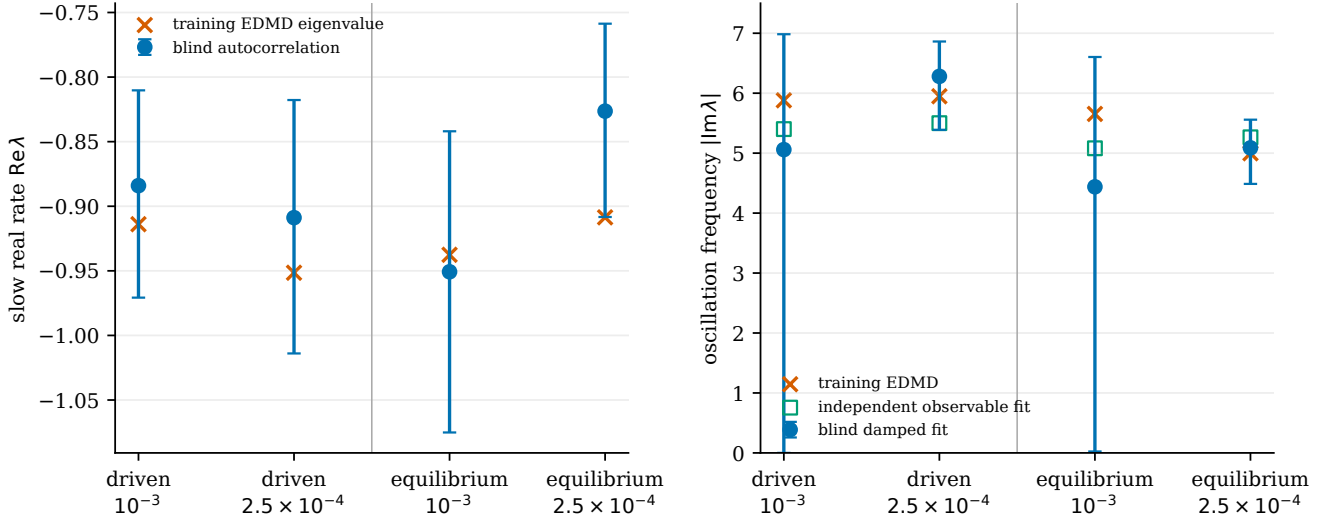


Figure 2: Training EDMD eigenvalues compared with independent blind-trajectory rate measurements. The coarse-step oscillatory intervals are broad; the fine step provides the useful precision.

Table 3: Held-out mode measurements. Residuals are test/validation.

Bath	dt	blind real rate [95% CI]	blind frequency [95% CI]	residuals	interpretation
driven	10^{-3}	-0.884 [-0.971, -0.810]	5.059 [0.0003,6.983]	0.807/0.850	coarse frequency broad
driven	2.5×10^{-4}	-0.909 [-1.014, -0.818]	6.281 [5.388,6.862]	0.668/0.733	resolved
equilibrium	10^{-3}	-0.951 [-1.075, -0.842]	4.438 [0.024,6.604]	0.751/0.794	coarse frequency broad
equilibrium	2.5×10^{-4}	-0.826 [-0.908, -0.759]	5.088 [4.487,5.558]	0.695/0.713	resolved

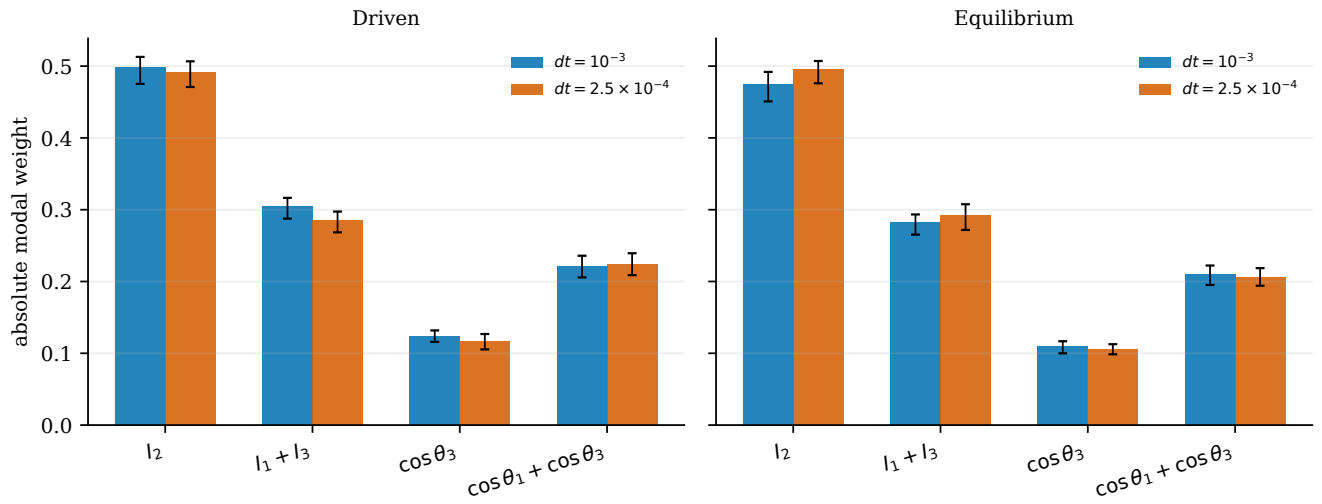


Figure 3: Absolute observable weights in the frozen slow real mode. Error bars are 95% whole-trajectory bootstrap intervals.